Excursions and Secular Variation of the Brunhes Epoch Geomagnetic Field in the Indian Ocean Region

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A total of 237 separate oriented samples have been taken from 33 separate lavas on Amsterdam Island, southeastern Indian Ocean. Six of these lavas, which are all of Brunhes age, record a 50°–80° latitudinal departure of the virtual geomagnetic pole from the present geographic pole. It appears that two departures, separate in time, recorded very similar virtual geomagnetic poles. Diminished geomagnetic intensities may have accompanied these departures, which could most readily be explained by a minor inclined dipole or nondipole feature persisting during a period of dipole collapse. The virtual geomagnetic poles for the remaining 27 lavas provide an angular standard deviation of 14.6° (with 95% confidence limits at 12.5° and 17.7°), which is consistent with present ratios of nondipole to main dipole fields in the latitudes involved. Comparison of the result with data from the Comore Islands (11.9°S) and Crozet (46.5°S) show that the Brunhes epoch geomagnetic secular variation in the Indian Ocean region was similar to the present geomagnetic field behavior.

In an earlier paper [*Watkins et al.*, 1972] paleomagnetic data from 12 lavas collected on Amsterdam Island (Figure 1) were incorporated into an analysis of Brunhes epoch ($t = 0$ to 0.69 m.y.) geomagnetic secular variation in the Indian Ocean and Atlantic Ocean regions. The reason for the small amount of data (representing two separate collections made in 1969 and 1970, respectively) is the great difficulty of landing on and surveying the island: visits can only be accomplished during the annual servicing of the station operated by the Administration of the Terres Australes et Antarctiques Francaises, and local weather frequently restricts movement on the island. In 1972 the first geological expedition to Amsterdam Island and St. Paul Island provided an opportunity to extend the sampling to a more satisfactory number of separate lavas: recent studies of Brunhes epoch geomagnetic secular variation [*Cox*, 1969*a*, 1971; *Doell*, 1970] employ from 22 to 31 separate sites, with at least 6 oriented specimens being taken from each site to facilitate adequate definition of the relevant remanent magnetism parameters [*Doell and Cox*, 1963]. Twenty-one additional igneous bodies were sampled during the 1972 survey. All units appear separate in time. Their locations are given, together with those of the earlier 12 units collected, in Figure 1.

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GEOLOGY OF AMSTERDAM ISLAND

Gunn et al. [1971] have described the geomorphology of Amsterdam Island (Figure 1) and have presented some geochemical data. The lavas analyzed have ranged from olivine tholeiites to high alumina basalts. The island appears to be in an early stage of evolution from oceanic ridge basalt to an alkaline volcano.

In outcrop, the island is readily divided into two contrasting divisions. An older volcanic episode is exposed in the very steep cliffs of the southwestern part of the island. In the central part of this cliff, a small peninsula composed of a shallow intrusive body crossed by narrow dikes of several trends, forms the oldest part of the island. Mont du Fernand (Figure 1) is a remnant of the rim of an old volcanic center, and a matching rim is preserved 1.7 km to the southeast. Draped around the old volcanic

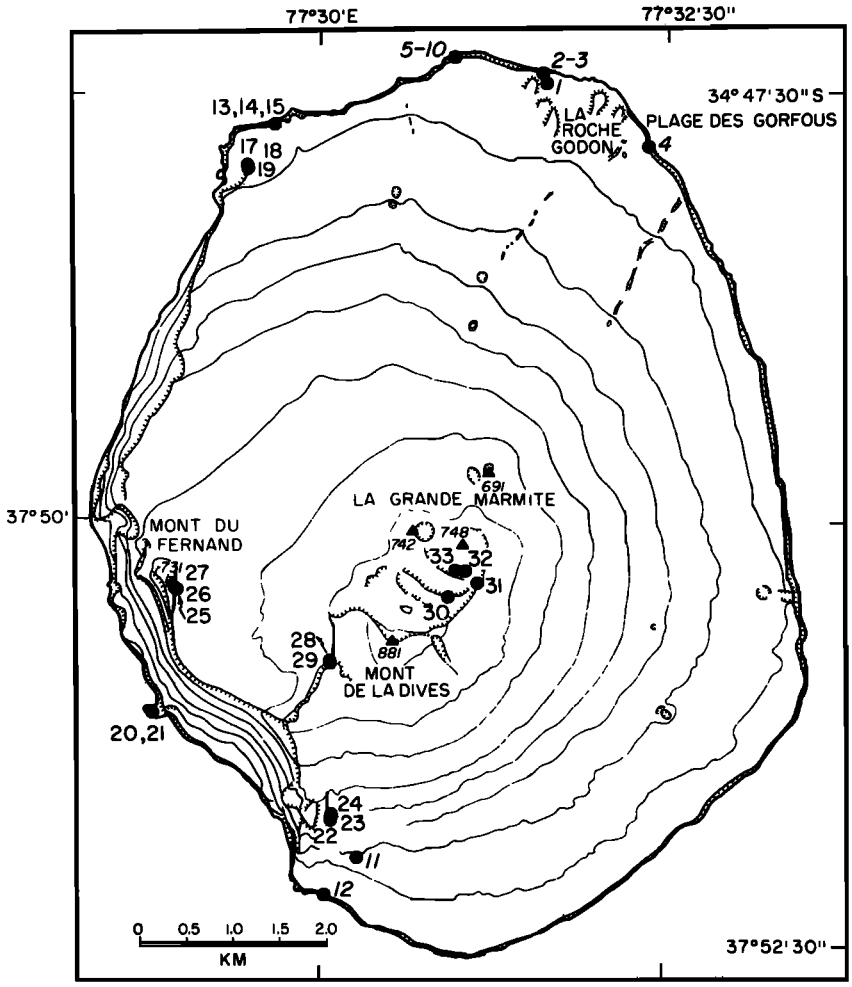


Fig. 1. Topographic map showing location of the sampled bodies on Amsterdam Island. Contours are in 100-meter intervals. Topographic features are shown as black triangles, with height given in meters. Steep cliffs are shown as hachure marks. Sample sites are shown as black dots, with numbers corresponding to those in Table 1.

center are the extrusives that constitute most of the island outcrop, providing a geomorphically young aspect. Superimposed on these draping lavas are several minor adventive cones, and craters at the island summit. The very restricted size of the island (and therefore the limited normal erosion potential) severely restricts confident interpretation of the landforms in terms of an approximate age estimate, but the island has all possible appearances of being restricted to the Brunhes epoch. It is probable that the older part of the island exceeds 0.1 m.y. in age.

METHODS

Field. A total of 237 cores of 2.5-cm diameter were collected from 31 bodies by using a gasoline-powered portable drill [*Doell and Cox*, 1965]. Cores were oriented geographically to a precision of ±3° while they were still attached to outcrop. Up to 5 lavas were taken in stratigraphic sequence. Sites 11 and 20–29 are from the geomorphically older parts of the island. Sites 30–33 are from the Recent summit crater. The remaining sites are from the lavas draping the old volcanic center.

Laboratory. A single specimen of 2.2-cm

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length was cut from each core. The remanent magnetism was measured by using a 5 cycles per second spinner magnetometer. Unstable components were minimized by demagnetization in alternating magnetic fields of 100 and 200 oe.

RESULTS

Results for all bodies are given in Table 1. The virtual geomagnetic pole (VGP) for each unit is shown in Figure 2. In Table 2, results are given for analyses of all sites combined.

DISCUSSION

Polarities. All bodies have VGP's in the northern hemisphere, or normal polarity. This confirms the assignment of the entire island outcrop to the Brunhes epoch on purely geomorphological grounds.

Departure of the geomagnetic field. Lavas 13, 17, 18, 19, and 24 possess VGP's in the vicinity of 280°E, 25°N (Figure 2). Lavas 17, 18, and 19 are in a single sequence on a scarp 0.5 km from the coastal cliff and may represent

TABLE 1. Paleomagnetic Data for Each Body

Site No.*	N	D	I	R	K	α_{95}	dp	dm	θ'	ϕ'
1	6	3.7	-58.6	5.996	240.3	2.29	2.1	2.8	86.7	197.0
2	8	70.4	-48.7	7.560	13.6	20.36	12.4	18.9	32.2	182.1
3	3	18.7	-66.6	2.990	381.7	1.66	3.4	4.2	72.3	213.9
4	8	25.3	-66.3	7.994	147.9	2.38	2.2	2.7	68.6	206.7
5	6	13.7	-32.3	5.971	171.3	6.18	3.3	5.8	66.6	112.1
6	6	14.0	-33.3	5.934	76.0	9.28	5.0	8.8	66.9	113.5
7	6	344.4	-48.9	5.974	189.4	5.87	4.3	6.4	74.8	14.0
8	6	4.1	-54.8	5.976	210.9	5.57	4.6	6.5	86.0	132.6
9	6	16.9	-57.6	5.971	173.2	6.14	5.5	7.5	76.7	175.1
10	6	18.8	-67.5	5.994	788.3	2.88	3.3	4.0	71.5	217.0
11	6	343.5	-30.8	4.928	4.7	38.11	21.6	38.8	64.4	38.5
12	8	345.1	-60.9	7.943	122.7	7.30	5.9	7.7	77.8	322.7
13	8	243.2	-68.2	7.991	808.6	1.95	2.8	3.3	14.9	292.7
14	6	19.8	-69.5	5.971	174.4	5.09	7.4	8.7	69.5	222.2
15	7	36.0	-64.2	6.989	531.4	2.62	3.3	4.2	62.2	196.0
16	8	6.2	-62.8	7.913	80.1	6.23	7.6	9.8	82.1	223.0
17	8	247.1	-76.2	7.985	464.3	2.57	4.4	4.8	24.5	284.0
18	8	247.6	-72.2	7.973	255.2	3.47	5.4	6.1	20.7	289.8
19	7	254.6	-81.7	6.910	66.5	7.46	14.0	14.5	31.9	276.2
20	8	15.6	-64.2	7.975	280.9	3.31	4.2	5.3	75.8	207.7
21	6	354.4	-57.0	5.993	682.1	2.57	2.7	3.7	85.6	348.9
22	6	357.9	-55.1	5.015	5.1	32.98	33.3	46.9	87.3	39.4
23	9	355.0	-57.1	8.898	78.6	5.84	6.2	8.5	86.0	348.2
24	9	257.1	-73.4	8.961	204.1	3.61	5.8	6.5	25.9	291.1
25	8	13.6	-62.9	7.986	503.7	2.47	3.0	3.9	77.8	204.6
26	8	352.4	-54.8	7.360	10.9	17.54	17.6	24.8	83.4	7.0
27	9	5.5	-60.7	8.822	44.9	7.77	9.0	11.9	84.2	212.0
28	8	330.7	-73.3	7.985	460.9	2.58	4.1	4.6	61.7	289.6
29	8	355.8	-62.3	7.978	323.0	3.09	3.7	4.8	83.4	285.4
30	7	3.2	-55.7	6.928	83.1	6.66	6.8	9.5	87.0	136.3
31	8	10.7	-61.9	7.988	578.5	2.30	2.8	3.6	80.3	204.3
32	8	358.1	-53.6	7.960	173.0	4.22	4.1	5.9	86.0	53.7
33	8	345.6	-51.0	7.920	87.1	5.97	5.5	8.1	76.7	10.7

*See Figure 1.

N, total number of separately oriented samples for given lava. D and I, mean declination and inclination of remanent magnetism, in degrees east of north and degrees below horizontal, respectively (obtained for unit vector per body). R, resultant vector (unit vector per body). $K = N - 1/N - R$. α_{95}, the semivertical angle of 95% confidence cone; dp and dm, major and minor axes of 95% confidence oval about mean dipole position, in degrees along and normal to great circle from mean sample site location to mean pole position, respectively. θ' in degrees north and ϕ' in degrees east, latitude and longitude of mean virtual geomagnetic pole obtained by using unit vector per body. All paleomagnetic data are for specimens treated in alternating demagnetizing fields of 100 or 200 oe.

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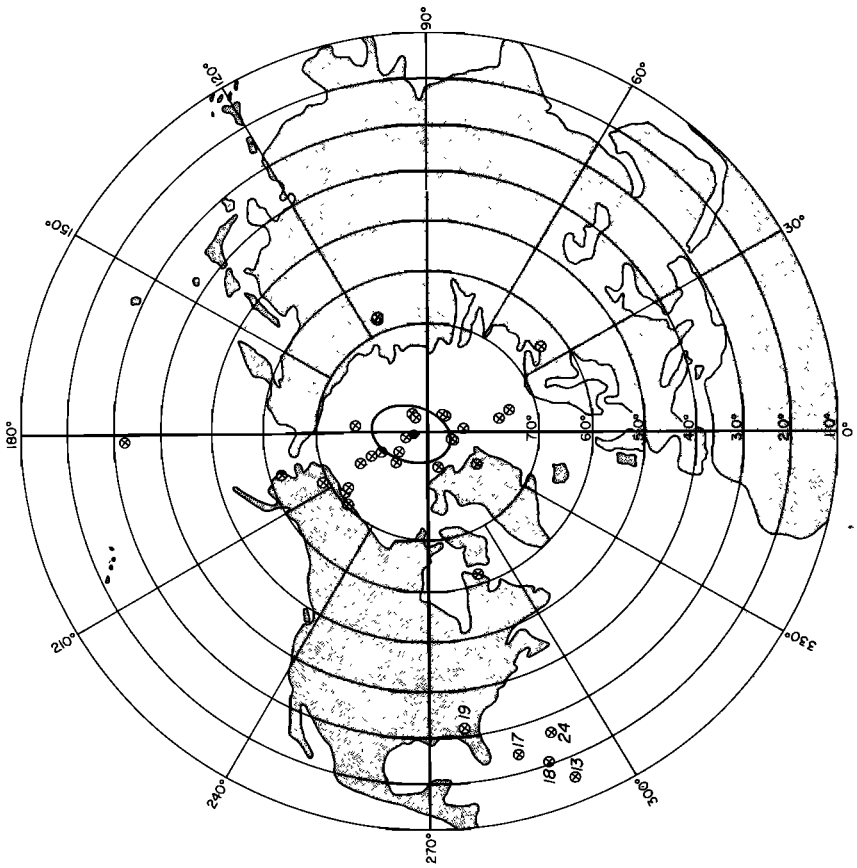


Fig. 2. Virtual geomagnetic pole positions for all sampled sites from Amsterdam Island (see Table 2 for exact data). The position of mean virtual geomagnetic pole and associated 95% confidence oval have been added. The site numbers (corresponding to numbers in Figure 1 and Table 1) are given for the low-latitude virtual geomagnetic poles only.

eruptions closely spaced in time. It is not impossible that lava 13, at the base of the coastal cliff, is from the same series of eruptions as lavas 17–19, but the two lavas (14 and 15) in the upper part of the cliff section both have high-latitude VGP's. Site 24, which also records the departure is, however, situated toward the base of a thick series of lavas assigned to the geomorphologically older series outcropping in ravines at the southern end of the island. The extreme geographic separation of site 24 from the lavas recording a very similar VGP, together with the great associated geomorphological contrasts, is evidence consistent with a departure of the geomagnetic field occurring with very similar characteristics at different times. If the same geomagnetic departure is represented in the two separate sections, then a drastic differential weathering would be required to explain the present geomorphology, and any estimates of the general volcanic stratigraphy of the island would be completely unable to utilize the strong geomorphological contrasts. The preferred interpretation of the present evidence is that two separate geomagnetic departures are recorded in the lavas of Amsterdam Island, but the absolutely unambiguous evidence for such a phenomenon, in the form of the separate geomagnetic departures represented by separated lavas in the same section, has not yet been obtained.

Doell and Cox [1972] have presented data from over 1000 separate Upper Tertiary lavas from various parts of the world showing that

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TABLE 2*a*. Statistical Analyses of Data, Southern Indian Ocean

Island	*B*	*N*	*D*	*I*	*R*	*K*	α_{95}	θ'	φ'	*dp*	*dm*
					Analysis of Directions						
Amsterdam*	33	237	0.6	-63.9	31.052	16.43	6.36	82.2	254.4	8.0	10.1
Amsterdam†	27	189	3.2	-58.0	26.329	38.72	4.52	87.3	188.1	4.9	6.7
					Analysis of VGP†						
Amsterdam	27				26.09	28.51	4.29	86.6	205.5		
Comores§	40				39.05	41.27	3.56	87.7	120.2		
Crozet§	79				74.91	19.06	3.75	86.8	121.3		

B, number of separate sites sampled. *N*, number of separate oriented samples. *D* and *I*, mean declination and inclination of remanent magnetism following demagnetization treatment and minimum dispersion computation; *D* is in degrees east of north, and *I* is in degrees with respect to horizontal (no sign means positive or below horizontal, and negative sign means above horizontal). *R*, resultant vector (unit vector per lava): $K = B - 1/B - R$. α_{95}, semivertical angle of 95% confidence cone. θ' and φ', latitude and longitude of virtual geomagnetic pole (VGP).

*For all data (Table 1).

†For θ' ≥ 45° only.

§See *Watkins et al.* [1972] for original data.

departures of VGP from the present geographic pole greater than 50° are recorded by only about 2% of the sampled units. No more than 7% of nearly 1000 separate Icelandic lava flows record VGP departures of more than 40° [*Dagley et al.*, 1967]. The phenomenon is clearly of short duration, perhaps lasting only about 10^3 years. The cause of such departures is best understood to be either nondipole activity during a period of diminished dipole intensity or behavior of the main dipole. The nondipole activity could be a polarity transition, or an 'aborted transition,' where the VGP returns to the original polarity after occupying the transitional low latitudes. *Doell and Cox* [1972, Figure 23] have defined three departures of varying but large amplitudes in Hawaiian lava flows. These departures follow great circles, which is strongly consistent with causes other than nondipole activity, the most reasonable

TABLE 2*b*. Statistical Analyses of Data, Southern Indian Ocean

Island	*S*	K_T	K_W	K_A	K_F	S_F	*Su*	*Si*
			*Computation of S_F with Respect to Mean VGP**					
Amsterdam	15.3	28.2	34.5	2100.0	32.4	14.2	17.2	12.1
Comores	12.7	40.8	39.2	2100.0	51.3	11.3	13.2	9.9
Crozet	18.7	18.7	55.2	2100.0	20.4	17.9	20.2	16.1
			Computation of S_F with Respect to Present Spin Axis					
Amsterdam	15.6	26.8	34.5	2100.0	30.6	14.6	17.7	12.5
Comores	12.9	39.6	39.2	2100.0	49.3	11.5	13.5	10.1
Crozet	19.0	18.0	55.2	2100.0	19.8	18.2	20.5	16.4

S, total dispersion; $S = (B - 1)^{-\frac{1}{2}} \Sigma_{i=1}^{B} (\delta i)^{\frac{1}{2}}$, where δi is the angular difference between the *i*th VGP and reference pole and *B* is the number of lavas. $K_T = 2(S)^{-2}$. K_W, the within-lava precision parameter for VGP; $K_W = \Sigma_{i=1}^{N} (Ni - 1)/(Ni - R_i)$, where *N* is the number of specimens in the *i*th lava and *R* is the resultant vector of the *i*th lava, applying the unit vector to the VGP for each specimen. K_A, the precision parameter for effect of magnetic anomaly existing during cooling of lava (see *Doell* [1972] for discussion). K_F, precision parameter reflecting ancient geomagnetic field variation, from $1/K_F = 1/K_T - 1/\bar{N}K_W - 1/K_A$, where $\bar{N}$ is the average number of cores per lava. $S_F = (2/K_F)^{\frac{1}{2}}$. *Su* and *Si* are upper and lower confidence parameters for S_F [after *Cox*, 1969*b*]. †All data are for bodies with θ' ≥ 45°.

*See Table 2*a*.

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explanation being a rapid tilting of the main dipole. The minimal nondipole activity in the central Pacific region for the past several million years [*Cox*, 1962; *Doell and Cox*, 1971, 1972] is compatible with this interpretation. *Barbetti and McElhinny* [1972] also reject secular variation as the cause of two geomagnetic departures of up to 120° of latitude, which were recorded by archeological materials in Australia between 28,000 and 31,000 years ago. Even if imperfections in the remanent magnetization acquisition and preservation are absent in the data involving other departures so far reported for the Brunhes epoch [*Smith and Foster*, 1969; *Bonhomment and Zähringer*, 1969; *Mörner et al.*, 1971], there is not as yet unequivocal support for polarity reversals existing during the Brunhes epoch [*Watkins*, 1971], despite bold assertions to the contrary [*Nature*, 1972].

Two lines of evidence can assist in resolution of the possibility that the departure recorded by the Amsterdam Island lavas was due to a tilting of the main dipole: (1) data from the same departure in other parts of the world, and (2) paleointensity results. If the former were available, then for main dipole behavior similar VGP's would be recorded at different locations on the earth's surface. Paleointensity measurements in the lavas extruded during a departure would, in the case of main dipole tilting, be expected to yield results consistent with a geomagnetic intensity which is not significantly lower than during periods represented by high-latitude VGP's. It is highly probable that the low intensities such as those recorded during polarity transitions [*Coe*, 1967; *Smith*, 1967*a*; *Goldstein and Strangway*, 1969] represent periods of very diminished dipole intensity, with a normal variable nondipole background.

None of the departures so far reported for the Brunhes epoch report VGP's in the vicinity of the Amsterdam Island results (Table 2, Figure 2). This fact, together with the lack of any evidence that a great circle movement of the geomagnetic field is involved, prohibits confident assignment of the data to main dipole movement. This interpretation is weak, however, because it depends on a lack of data rather than on any positive evidence.

In Figure 3 the average intensity of magnetization of the Amsterdam Island lavas is plotted against the associated VGP latitudes. *Lawley* [1970] and *Dagley and Wilson* [1971] have shown convincingly that for large amounts of data from Icelandic basalts, systematically diminished average intensities of magnetization accompany low-latitude VGP's, suggesting that the rigorous experimental methods normally required for relative (if not absolute) paleointensity determinations on basalts [*Smith*, 1967*b*] may be substantially augmented by large amounts of simple intensity of magnetization data, however large the within-lava variation of this parameter may be [*Watkins and Haggerty*, 1967; *Wilson et al.*, 1968]. Although the results from Amsterdam Island are limited (237 samples from 33 lavas), those lavas recording the departures nevertheless possess the lowest intensities of magnetization (Figure 3), which is quite consistent with the Icelandic results. This result is easiest to attribute to a diminished pole intensity during the geomagnetic departure.

If nondipole activity is the controlling factor in the position of the observed low-latitude VGP's, the preferred interpretation of the data in terms of two very similar departures that are separate in time is subject to obvious difficulties. *Creer and Ispir* [1970] believe that a quasi-permanent minor inclined dipole may be a feature of part of the Tertiary period, as is evidenced by similar VGP paths being recorded paleomagnetically during several polarity transitions. The existence of such a feature could explain repeated VGP's during different geomagnetic departures. Although there is no evidence as yet that the Amsterdam Island VGP's are associated with such an inclined dipole, at the same time (as was argued earlier) all field evidence is against contemporaneity of unit 24 and the lavas to the north that recorded such similar low-latitude VGP's. To explain some Japanese paleomagnetic data, *Creer and Ispir* [1970, pp. 289–290] have formulated a geomagnetic field model, one of the properties of which is a geomagnetic pole near that of the VGP for lava numbers 13, 17, 18, 19, and 24 (Figure 2). Since the rocks involved are Miocene in age, however, this model cannot be considered as relevant to the Brunhes epoch results presented here. A second possibility is that the same nondipole feature persisted during both periods of

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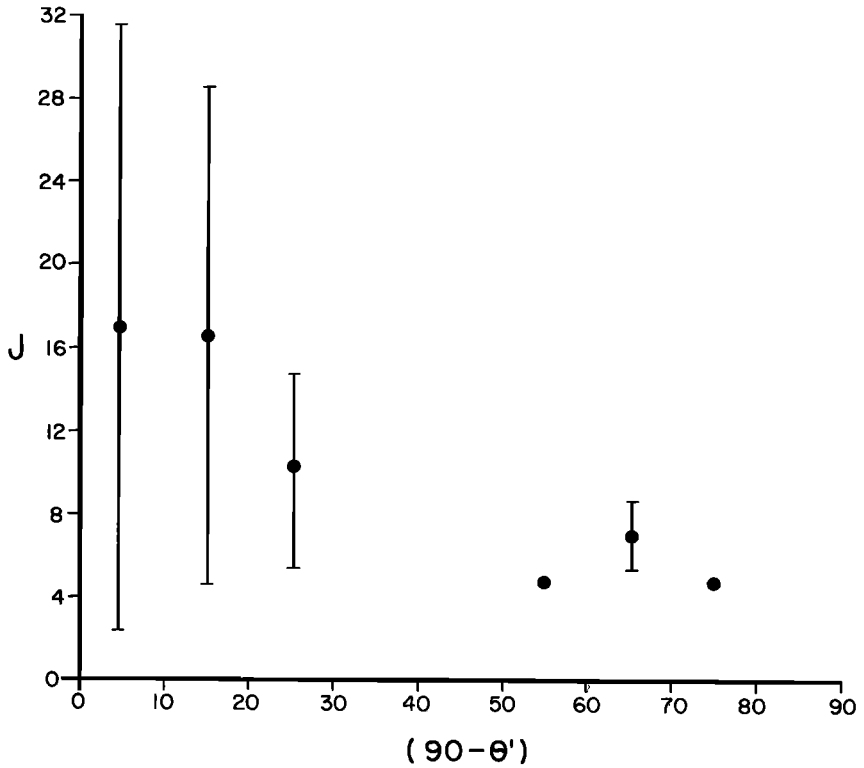


Fig. 3. Mean intensity of magnetization J in emu/g $\times 10^4$ for each separate body (Figure 1 and Table 1) as a function of paleomagnetic colatitude ($90° - \theta'$), for 10° increments. The standard deviation of J is added when there is more than one data point per 10° increment. See Table 1 for paleomagnetic latitude θ' of each body.

diminishing main dipole intensity. This would require reevaluation of the concept of the rapidly moving and changing nondipole field [*Honkura and Rikitake*, 1972].

Secular variation. The data have been analyzed by using the methods of *Cox* [1969*a*, *b*, 1971] and *Doell* [1970], in which the total VGP angular standard deviation S is corrected for within-lava precision K_W and for the probable effect on the precision of the magnetic anomaly existing during the cooling of the lavas K_A, to arrive at a best estimate of the ancient geomagnetic field dispersion S_F, which is then related to latitude θ for comparison with the various geomagnetic field models: model A, in which a vector of varying orientation but constant amplitude (to simulate nondipole activity) is added to a fixed axial dipole [*Irving and Ward*, 1964]; model B, in which the geomagnetic field is envisaged as being due to wobble of the main dipole only [*Creer et al.*, 1959]; and model C, which combines dipole wobble and a randomly varying nondipole component [*Cox*, 1969*a*]. Values of S_F that can be obtained from the Amsterdam Island results are given in Table 2. The results reflecting the geomagnetic excursion discussed above have not been included in this analysis, since they clearly reflect a behavior that is anomalous. Figure 4 shows S_F, together with the 95% confidence limits [*Cox*, 1969*b*], superimposed on the S_F versus latitude relationships corresponding to the various geomagnetic field models. Added to Table 2 and Figure 4 are the only other results from the Indian Ocean region for the Brunhes epoch, which use high-density sampling. The method in the field and laboratory and the computation methods are exactly the same as those used in the Amsterdam Island study.

The value of S_F obtained from the collection of 27 separate lavas (Table 2) contrasts favorably with the preliminary results [*Watkins et*

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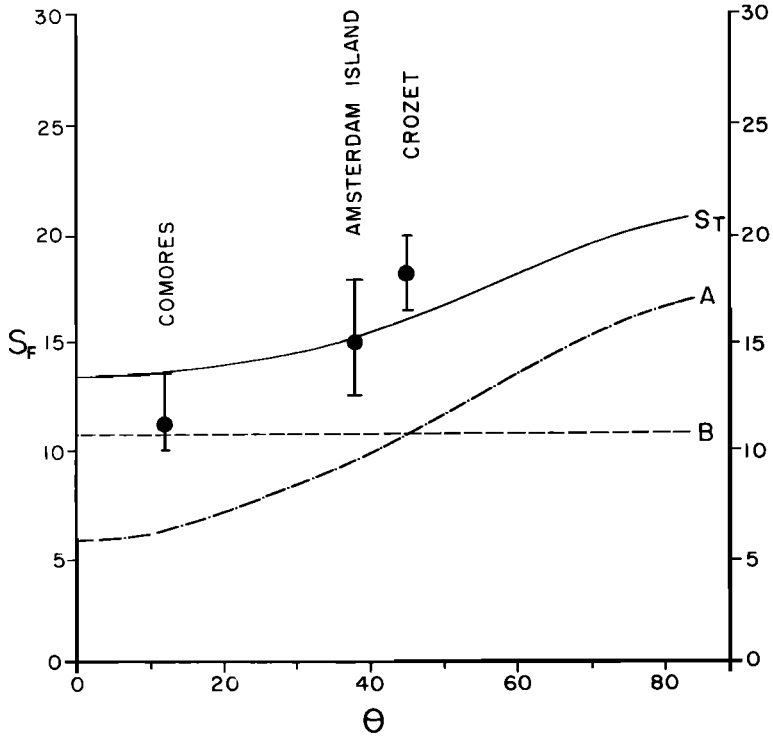


Fig. 4. Relationship between angular standard deviation of the virtual geomagnetic poles for all data (S_F) and geographic latitude (θ). See footnote to Table 2*b* for definition of S_F and text for geomagnetic field models corresponding to A and B. Effect of total field behavior of the 1964 International Geomagnetic Reference Field is denoted by S_T. Value for B is from *Doell* [1969]. Data from the Comores and Crozet islands are published elsewhere [*Watkins et al.*, 1972] but are presented here in form of S_F values for the first time.

al., 1972], which were based on 12 lavas only. This earlier survey yielded a value of $S = 24.4$ with upper and lower 95% confidence limits at 35.3 and 18.8, respectively, which is significantly larger than the final result reported here. This is evidence that insufficient sampling of the geomagnetic time variations was made during the initial survey.

The results in Figure 4 provide two conclusions: (1) as is shown in earlier studies, the dipole wobble model is excluded; and (2) the Brunhes epoch geomagnetic secular variation in the southern Indian Ocean region was not dissimilar to that of the present geomagnetic field, although a more rapid increase of S_F with latitude may have existed. The data remove the possibility of any substantial attenuation of nondipole field activity such as that which existed in the central Pacific Ocean during the same period [*Doell and Cox*, 1972]: this places restrictions on the geographic limits and possible movement of the lateral inhomogeneity in the lower mantle, which *Cox and Doell* [1964] and *Doell and Cox* [1972] believe best explains the Pacific Ocean paleomagnetic secular variation minimum.

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